$F = 400\ \text{lbf}$ $\qquad \delta = 0.009\ \text{in}$

$E = 10 \cdot 10^6\ \text{psi}$ $\qquad w = 0.75\ \text{in} \qquad h = 0.5\ \text{in}$

$$A = wh = 0.75 \cdot 0.5 = 0.375\ \text{in}^2$$

$$\delta = \frac{FL}{AE} \rightarrow L = \frac{AE\delta}{F} = \frac{(0.009 \cdot 0.375 \cdot 10000000)}{400}$$

$$L = 84.375\ \text{in}$$

$$\sigma = \frac{F}{A} = \frac{400}{0.375} = 1066.67\ \text{psi} = 1.07\ \text{ksi}$$